



FIRST SEMESTER 2026-2027

Course Handout Part II

Date: 01/08/2026

In addition to part-I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No. : DE G611
Course Title : DYNAMICS & VIBRATION
Instructor-in-charge : Dr. Arun Kumar Jalan

1. Course Description:

Steady and transient Vibration of single and multi degree freedom systems. Systems with distributed mass and elasticity. Non-linear and self-excited vibrations, structural damping, Random vibrations, vibration analysis, vibration control - reduction, isolation and vibration absorbers.

2. Course Objectives:

No	Course Objectives
CO1	Provide an introduction to analytical dynamics and present Lagrange's equations of motion which is an efficient way of deriving equations of motion. This also is a prerequisite for later content of the course.
CO2	Present a matrix approach to the vibration of multi degree of freedom systems, placing emphasis on modal analysis. To cover methods suited for automatic computation for obtaining the system response.
CO3	Setting-up initial-boundary value problems for some important and fundamental structural members viz. Bars, strings, rods and beams. Analytical and approximate solutions to these problems for various loading and boundary conditions are discussed and analyzed.
CO4	Learn to use finite element methods for the analysis of the vibrations of structures.

3. Learning Outcomes

No	Learning Outcomes
LO1	An ability to apply mathematical knowledge to an engineering problem by developing the equations of motion for vibratory systems and solving them.
LO2	Understanding importance of how structural vibrations may affect safety and reliability of engineering systems.
LO3	An ability to use finite-element methods to analyze structural vibrations
LO4	An ability to use modern engineering tools necessary for engineering practice such as MATLAB.

4. Text Book:

"Elements of Vibration analysis", Leonard Meirovitch, McGraw-Hill, Singapore, 1986.





5. Reference Books:

- "Vibration Theory and application", William T. Thomson, CBS Publications, 3rd Ed., 1988.
- "Mechanical Vibrations - Theory and Application" Francis S. Tse, Ivan E. Morse and Rolland T.Hinkle, Allyn and Bacon Inc. London, 1983.
- "Advanced Engineering Dynamics", Harrison and Nettleton, John Wiley & sons. New York. 1997.

6. Course Plan:-

PART (A)

Lect. No	Topics to be covered	Learning Objective	Reference Chap./Sec
1-2	Review Newtonian Mechanics	Newton's law, Principle of virtual work, D'Alembert's Principle	RB (C): 1
3	Lagrange Mechanics Formulation	Lagrange Mechanics	RB (C): 2
4-7	Application of Lagrange Mechanics	Particle and rigid body problems	RB (C): 2

PART B

8-15	Introduction to Vibration	Review of Single DOF free & forced Vibrations(steady state and transient)	TB : 1 RB (B) : 1,2
16-18	Understanding higher dof concept	Two DOF, Modal vectors Eigen value Problems	TB : 4
19-20	Extension to Multi dof	Lumped Parameter model of continuous system	TB : 4
21-22	Problems in Multi-DOF	Problem solving	TB : 4
23-25	Approximate solutions Continued	Holzer's method	TB : 7
26-27	General Discussion of continuous Systems	Continuous system Exact Solution	TB : 5
28-29	Continuous models	String problem, Axial vibration of rod	TB : 5
30-31	Continuous Model: Exact Solutions	Bending vibration of Bars Boundary conditions	TB : 5
32-34	Continuous system: Approximate solutions	Rayleigh's energy method, Rayleigh-Ritz method	TB : 7
35	Finite Element method	General Discussion, Stiffness Matrix by direct approach	TB : 8
36-37	Equation of motion using FEM	Assembly process	TB : 8
38-39	Solution of equation of motion (FEA)	Eigen value problem, Rayleigh-Ritz method	TB : 8
40	Introduction to Nonlinear systems	General consideration, Perturbation Technique	TB : 10

7. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Remarks
Mid-sem Exam	90 min	25	<TEST_1>	CB
Quiz / Test		10		CB
Project + Seminar		30	-	OB
Comprehensive Examination	180 min	35	<TEST_C>	CB + OB

8. Consultation Hours: Chamber no 2229 Wednesday 4.00 pm to 5.30 pm

9. Notice: All the notices regarding the course will be displayed on Nalanda or, ME Department Notice Board.

Instructor-in-charge
DE G611

